

What Is a Persona Worth?

Pricing LLM Human Simulation in Human Answers

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Abstract

An LLM persona is supposed to stand in for a person. We make that claim measurable by pricing it in the only currency a practitioner actually spends: questions asked of a real human. The *answer-equivalent* m^* of a simulator is the number of a respondent's own answers a conditional model needs before it predicts that respondent's remaining answers as well as the simulator does. It is defined against a strictly proper ordinal score, it is selected entirely on held-out respondents, and it does not presuppose its own answer: $m^*=40$ would be strong evidence that persona prompting substitutes for a questionnaire. We first show why a proper score is required at all — under the answer-similarity metric these systems are validated with, a simulator that reproduces the target population's response distribution *exactly* can never beat a constant vector, and is strictly beaten on all 480 cluster-item combinations of 410,168 IPIP-NEO-120 respondents. We then price 72 (model $\times$ instrument $\times$ panel) cells spanning four inventories and fifteen open models, including a held-out family of 180 IPIP-NEO-300 items the same respondents answered. The answer-equivalent is stable across disjoint respondent panels (Spearman $\rho=0.78$, $p<10^{-6}$) and orders models consistently: the strongest persona we measure is worth 5.38 of 120 answers on the unseen item family and 3.07 of 60 on the short form, the weakest exactly 0. What separates them is not accuracy. It is the share of each model's person-specific deviation that survives population calibration ($\rho=0.77$ with m^* ; raw accuracy does not predict m^* once that is controlled), and how the observations are presented: a 30-line trait profile is worth almost five times the raw transcript of the same 120 answers. Finally we bound the result: across 160 cells and three combination families — convex, linear, and gradient-boosted — adding a persona to a conditional model of the respondent's own observed answers yields a selected weight of 0 in 127 cells and never above 0.05;

of 426 tests one survives multiplicity control, and it reverses sign on its own disjoint panel. A persona carries a small, real, model-scaling amount of personal information that a handful of real answers renders redundant. We release the instrument, the code and all per-cell results.

1 Introduction

A practitioner deciding whether to simulate respondents instead of recruiting them is asking a quantitative question: how many real answers is the simulator worth? The literature does not answer it. Simulators are reported as correlations, agreement scores, or wins over a baseline, none of which convert into the resource the decision actually trades against — questions put to a real person.

We supply that conversion. The *answer-equivalent* m^* of a simulator is the number of a respondent's own answers a conditional model needs before it predicts that respondent's remaining answers as well as the simulator does.

Two properties make it usable. First, it is anchored to a strictly proper score, which is not optional here: the answer-similarity metric these systems are validated with has a degenerate optimum, and under it a simulator that reproduces the target population's response distribution *exactly* is dominated by a constant vector (§3). A price quoted in an improper currency is not a price. Second, it does not presuppose its answer. Every quantity is selected on held-out respondents, and $m^*=40$ would be strong evidence that persona prompting substitutes for a questionnaire. That symmetry is what makes the number we do measure worth reporting.

Our contributions are:

1. **The answer-equivalent** (§10): an instrument that prices any respondent simulator in units of the respondent's own answers, under a strictly proper ordinal score, with every hyperparameter selected on held-out respondents

and the baseline’s capacity returned at each budget.

2. **Why a proper score is required (§3):** under answer-similarity, an exactly faithful simulator is dominated by the best constant predictor — strictly so on all 480 cluster–item combinations we measure. The penalty is the excess of the Gini mean difference over the mean absolute deviation about the median, divided by the scale range.
3. **A price, at scale (§11):** 89 cells over five inventories and fifteen open models, including transfer to 180 unseen items the same people answered. m^* replicates across disjoint respondent panels ($\rho=0.78$) and reaches 5.38 answers where human labels are scarcest.
4. **A mechanism (§12):** m^* is governed by the share of a model’s person-specific deviation that survives population calibration, not by its accuracy.
5. **A boundary (§13):** across 160 cells and three combination families, a persona adds no predictive information beyond a conditional model of the respondent’s own observed answers; the selected combination weight is 0 in 127 cells and never exceeds 0.05.

2 Setting

Task. A population of respondents is partitioned into K personality clusters by k -means over the 30 IPIP-NEO facet scores. For each cluster a natural-language description is used to condition an LLM, which then answers the 120 IPIP-NEO items on a 1–5 Likert scale. Prompt wording is optimised by an evolutionary search whose fitness is answer agreement with held-out human members of the cluster.

Two regimes appear here and are not interchangeable. The *cluster* regime conditions on the centroid alone, giving every member one prompt; it is what Prop. 1 and the 7.75-point penalty address. The *individual* regime, used by the released prompt builder and by §14, inserts each respondent’s own scores as intensity modifiers. By Remark 1 one constant is optimal in the first and a per-respondent conditional median in the second; we name the regime wherever a result depends on it.

The metric. A simulated answer vector x is scored against a human answer vector y by

$$S(x, y) = \frac{1}{J} \sum_{j=1}^J \left(1 - \frac{|x_j - y_j|}{4} \right), \quad (1)$$

with $J=120$ and the divisor 4 the scale range. Cell scores average S over the held-out respondents. Equation (1), or a monotone transform of it, is the objective the search optimises and the number reported.

3 A constant is optimal under S

Fix a cluster and an item j , and let P_j be the cluster’s response distribution over $\{1, \dots, 5\}$. Write $Y \sim P_j$ for a randomly drawn human respondent.

Proposition 1. *Let m_j be a median of P_j , let $\text{MAD}_j = \mathbb{E}_Y |m_j - Y|$ and let $\text{GMD}_j = \mathbb{E}_{X,Y} |X - Y|$ for $X \sim P_j$ independent of Y (the Gini mean difference). Then*

- (a) *the best constant predictor attains expected score $1 - \text{MAD}_j/4$;*
- (b) *a faithful simulator, i.e. one whose output is distributed as P_j , attains expected score $1 - \text{GMD}_j/4$;*
- (c) *$\text{GMD}_j \geq \text{MAD}_j$, with equality iff X lies almost surely in the median set of P_j .*

Hence the constant is at least as good as the faithful simulator, and strictly better unless P_j places all its mass on medians.

Remark 1 (Where this applies). *Proposition 1 is unconditional; the audited systems condition on something. Let C be everything the simulator observes, its draw independent of the response given C . Under (1) an optimal simulator emits a conditional median $m_j(C)$, so one constant stays optimal exactly when the conditional median sets share a value almost surely, item by item. Cluster-only conditioning thus admits a cluster-specific constant optimum; individually informative conditioning can make different deterministic answers optimal, so the constant need not win. A deterministic conditional median is always optimal and a randomised predictor ties exactly when its support lies in that median set, so absolute error never uniquely rewards faithful conditional sampling; richer conditioning moves which prediction is rewarded.*

Proof. (a) and (b) are immediate from (1). For (c),

$$\begin{aligned} \text{GMD}_j &= \mathbb{E}_X [\mathbb{E}_Y |X - Y|] \\ &\geq \mathbb{E}_X [\min_c \mathbb{E}_Y |c - Y|] \\ &= \mathbb{E}_Y |m_j - Y| = \text{MAD}_j, \end{aligned}$$

since the median minimises expected absolute deviation. Equality requires X to attain that minimum almost surely, i.e. to lie in the median set of P_j . This is weaker than degeneracy: a balanced two-point law $P(Y=1) = P(Y=5) = \frac{1}{2}$ has $\text{GMD} = \text{MAD} = 2$, so the bound is tight there despite large variance. On the IPIP items it is strict throughout (Table 2). $\square$

The consequence is not a technicality. Under S the behaviour we want — reproducing the population’s response distribution — is penalised relative to one fixed vector, so the optimum is degenerate in exactly the variance that carries a profile. Whether an optimiser is driven there is empirical and not uniform: S_0 ships a near-worst model from a heterogeneous pool (App. D), yet prompt search under it improves proper scores too (App. E).

3.1 The metric is CRPS with its dispersion term deleted

Proposition 1 is not an accident of Likert data. The continuous ranked probability score for a predictive distribution F and outcome y is

$$\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|,$$

and it is *strictly proper*: uniquely optimised by $F = P$. Equation (1) is its first term alone. Deleting the second is exactly what makes a constant optimal.

This yields the natural repair. Define

$$S_\lambda = 1 - [\mathbb{E}|X - y| - \lambda \mathbb{E}|X - X'|]/4, \quad (2)$$

$\lambda=0$ is the audited metric — improper, constant-optimal. $\lambda=1/2$ is CRPS — strictly proper, truth-optimal. $\lambda>1/2$ is improper in the opposite direction. $\lambda=1/2$ is the unique proper choice, not a tunable hyperparameter.

Sweeping λ over the same 20 cells (Table 1) shows the repair works and what it costs the simulator. At $\lambda=1/2$ the human ceiling (0.859) rises decisively above the constant floor (0.796): the ordering that §7 shows is inverted under the audited metric is *correct* under the proper one. Humans overtake the constant at $\lambda^*=0.30$, comfortably below the proper coefficient. The model needs

λ	Model	Constant	Human
0 (audited)	0.726	0.796	0.720
0.25	0.749	0.796	0.789
0.5 (CRPS, proper)	0.772	0.796	0.859
0.75	0.795	0.796	0.929
1.0	0.819	0.796	0.998

Table 1: The S_λ family of (2), averaged over 20 cells, all values on the corrected orientation. A constant wins at the audited $\lambda=0$; at the proper $\lambda=1/2$ a real cluster member wins, as it must. All-pairs scores, slightly below the paired ones of §4; the ordering is the same under either.

Clu.	n	MAD	GMD	S_{const}	S_{faith}
0	107,174	0.804	1.107	0.7990	0.7232
1	108,880	0.723	1.008	0.8192	0.7480
2	108,867	0.809	1.128	0.7978	0.7181
3	85,247	0.871	1.204	0.7822	0.6990
Mean		0.802	1.112	0.7996	0.7221

Table 2: Proposition 1 on 410,168 IPIP-NEO-120 respondents. $S(\text{const})$ is the best constant predictor; $S(\text{faithful})$ is a simulator that reproduces the cluster response distribution exactly. The faithful simulator is worse in every cluster.

$\lambda^*=0.75$ (median grid crossover; 0.757 on the aggregate curve) — a rule that over-rewards dispersion — and beats the constant in only 2 of 20 cells at the proper $\lambda=1/2$, though it does overtake in most cells once λ reaches 1, where the reward for dispersion is twice proper. So the simulator’s failure is not an artefact of a bad metric: it survives the correct one.

so larger is better. Scores in the appendices are reported as the corresponding $\text{loss } L_{1/2} = 1 - S_{1/2}$, the normalised CRPS, where smaller is better; we use the loss wherever a table compares errors and mark the direction in each caption.

Table 2 evaluates the bound on the real data: the structural penalty against a faithful simulator averages **7.75 points**, and $\text{GMD}_j \geq \text{MAD}_j$ holds on 120/120 items in all four clusters.

4 Empirical audit

Setup. We re-score an existing cluster-level digital-twin system on its own artefacts: the same 410,168-respondent IPIP-NEO-120 corpus, the same $k=4$ clusters, and the exact train/test respondent IDs it recorded, for five LLMs (GigaChat3-10B, GPT-4.1-mini, GPT-4.1-nano, Grok-4.1-fast, Qwen3-235B) $\times$ four clusters. We add no LLM calls: model answers are read from the released run

artefacts. Baselines are fitted on the recorded train IDs and scored on the recorded test IDs.

Baselines. *Cluster mean*: the per-item rounded mean of the cluster’s training respondents, one constant vector. *Cluster majority*: the per-item mode. *Global mean*: the per-item grand mean over all respondents, ignoring clusters. *Empirical sample*: sample each item independently from the cluster’s training response distribution. *Real human*: copy the answer vector of a randomly chosen training respondent from the same cluster.

Result. Table 3 reports S . The evolutionary persona averages 0.660 as released and 0.739 after the orientation correction of §6; the constant cluster mean averages 0.796. The LLM loses in 20 of 20 cells under either orientation, by 13.6 and 5.8 points respectively. What survives correction is *matched-cell* dominance: the model loses to its own cluster’s constant in every cell, while its best cell (0.807) now exceeds the weakest constant elsewhere (0.779), so the comparison only holds within a cluster. A per-item global mean — which never looks at the cluster at all — scores 0.774, still above the persona’s corrected 0.739.

5 Does conditioning on the right cluster matter?

Correct cluster assignment does help, and by far too little to matter: the permuted-cluster control is worse than the correct one, but both are worse than a constant. Appendix R gives the per-cell numbers. The individual-level version of this control — correct respondent versus shuffled respondent — is what §10 prices, and it is the version that bears on whether a persona simulates a *person*.

6 An orientation defect, and what survives it

The released benchmark scores raw model self-reports against reverse-recoded human answers on 55 of 120 items. Correcting it lifts the correlation of per-item means from +0.09 to +0.63 and the model’s score from 0.660 to 0.739, and leaves every conclusion above intact: the constant still wins 20 of 20 cells. We apply the correction throughout, including to the persona readouts priced in §11, where 43 of 60 IPIP targets are reverse-keyed and omitting it would silently corrupt every simulator arm. Appendix T gives the audit in full.

7 The human ceiling sits below the constant floor

The decisive check on the metric is what a *real* member of the target cluster scores, since a cluster-level twin cannot do better than being an actual member. A randomly drawn same-cluster respondent scores 0.721 and is beaten by the constant cluster mean in 20 of 20 cells. A metric that ranks a constant above a genuine group member is not measuring simulation, which is why the price in §10 is anchored to a proper score instead. Appendix W gives the per-cell numbers and the bootstrap.

8 A replacement suite

A usable metric must (i) be maximised by distributional fidelity rather than central tendency, and (ii) be reported against an explicit *human ceiling* and *constant floor*, so a number can be read as progress toward a real target rather than in the abstract. We propose three, computed from a set of simulated respondents rather than from paired individuals.

C2ST-AUC (primary). Train a gradient-boosted classifier to separate n simulated answer vectors from n held-out human vectors of the same cluster; report cross-validated AUC, folded to $\max(\text{AUC}, 1 - \text{AUC})$. An AUC near 0.5 means this classifier cannot separate the samples at this sample size — evidence of indistinguishability under the stated configuration, not of distributional equality (App. N). Unlike S it cannot be gamed by collapsing onto the mean.

Distributional fidelity. $1 - \frac{1}{J} \sum_j W_1(\hat{P}_j^{\text{model}}, \hat{P}_j^{\text{human}})/4$, with W_1 the per-item Wasserstein-1 distance. Rewards matching the full response distribution.

Variance ratio. $\frac{1}{J} \sum_j \text{sd}(\text{model}_j)/\text{sd}(\text{human}_j)$; 1.0 is ideal, a constant gives 0. A diagnostic, not an optimisation target.

Results. Table 4 applies the suite to the same 20 cells. All three order human > constant, as required; the old metric does the reverse. Two substantive findings follow. First, current personas are *near-perfectly separable* from real humans (C2ST-AUC 0.994 against a human ceiling of 0.564): a classifier separates a persona from the group it simulates almost perfectly, while barely separating one human sample from another. The constant scores 0.926 here, but we do not read 0.994 against 0.926 as an intrinsic ordering — constant detection

Model	Cluster				Mean
	0	1	2	3	
GPT-4.1-mini	0.657	0.638	0.710	0.652	0.664
GPT-4.1-nano	0.673	0.678	0.697	0.670	0.679
GigaChat3-10B	0.711	0.691	0.737	0.672	0.703
Grok-4.1-fast	0.598	0.583	0.659	0.614	0.613
Qwen3-235B	0.634	0.615	0.677	0.642	0.642
<i>Cluster mean (constant)</i>	0.794	0.827	0.785	0.779	0.796
<i>Real human, same cluster</i>	0.720	0.747	0.715	0.704	0.721

Table 3: Answer similarity S on the original system’s own test IDs, in the *original* orientation; correction adds 0.079 on average and preserves all 20 matched constant wins, though it does reorder the models. The constant (bold) beats every LLM in every cell; a real human scores below it.

reaches 1.000 with more data, so the gap between two already-detectable systems is a property of the classifier. Second, they are systematically under-dispersed: $VR = 0.43$, a ratio of standard deviations, so LLM personas produce well under half the response *spread* of the humans they simulate (Figure 1). Evolutionary optimisation moves none of these appreciably.

Metric	LLM	Human ceiling	Const. floor
C2ST-AUC ↓	0.994	0.564	0.926
DF ↑	0.783	0.939	0.795
VR (1.0 ideal)	0.431	1.018	0.000
$S_{\text{answer}} \uparrow$ (old)	0.739	0.721	0.796 †

Table 4: The suite over 20 cluster $\times$ model cells, each metric against its human ceiling and constant floor. †: under the old metric the constant floor *exceeds* the human ceiling — the failure this paper documents.

Under-dispersion is the wrong summary. A single variance ratio hides which variance is missing. Writing each answer matrix as $Y_{pj} = \mu + a_p + b_j + r_{pj}$ — respondent main effect, item main effect, and the two-way residual — tells a sharper story. One observation per person–item cell cannot separate a stable interaction from response noise, so we read this term as a residual, not as individuation. The simulator emits nearly the *right total* variance (ratio 0.942 over the 20 cells) but places it differently: the item main effect takes 81.2% of its variance against 29.8% for humans, while the residual takes 17.7% against 67.0%. The residual SD ratio is 0.466, and 0 for a constant by construction.

So the simulator is not emitting too little. It is emitting a confident cluster-level *answer key* — which items are endorsable — with little variation in which *person* endorses which item. We recommend reporting the residual ratio alongside the plain variance ratio, since the two rank systems differently. Individuation itself we measure with the aligned amplitude of §14 and the correct-versus-permuted CRPS, which noise cannot inflate: permutation leaves the residual unchanged.

The style-only rung of the C2ST ladder, and what separability it carries, is reported in Appendix Q.

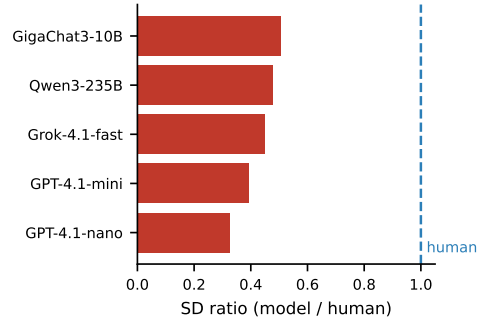


Figure 1: Response *spread* relative to the humans being simulated, as a ratio of standard deviations. Every model is substantially under-dispersed.

9 The same penalty, in a fitted decoder

The proposition concerns a faithful sampler against a constant. A fitted predictor shows the same pressure. Fitting on half of each cluster and scoring on the other half, two oracles differ only in what they emit: a **mean** decoder (ridge on the ordinal code) predicting the conditional mean, and a **residual-resampled** one, the same fit plus a resampled residual, which restores marginal spread. The mean decoder scores $S = 0.802$, the residual-resampled one $S = 0.730$ — worse by 7.2 points for being closer to the response distribution on that margin. This is an illustration, not a test of Prop. 1: the

decoder conditions on a richer code than the belief prompts carry, so its 7.2 points are not a prediction of the GMD – MAD gap. It beats the constant (0.792) by 0.9 points — what one fitted reader achieved, not a bound (App. W).

10 The answer-equivalent

Definition. Let S be a strictly proper ordinal score — we use the ranked probability score, normalised by the $K-1$ thresholds, so S is comparable across the 5- and 7-point inventories we study. Let a respondent i have observed answers x_i and held-out answers y_i on disjoint item sets. Write D_m for a conditional model fitted on training respondents that sees m of respondent i ’s observed answers, and $\hat{F}$ for the simulator under test. The answer-equivalent is

$$m^*(\hat{F}) = \min\{m : \mathbb{E} S(D_m, y) \leq \mathbb{E} S(\hat{F}, y)\}.$$

We report both this integer crossing, which is what a practitioner acts on, and its linear interpolation between the bracketing budgets, which is the honest effect size: the crossing is coarse and rounds a simulator up by as much as a whole answer.

What the simulator is allowed. $\hat{F}$ is the persona readout after population calibration; the raw readout scores worse than the population prior, so pricing it raw would price the prompt format rather than the model. We fit $\hat{F}_{ij} = \Pi_{\text{CDF}}[F_{0j} + \alpha(Q_{ij} - \bar{Q}_j)]$, with Q the model’s own CDF from its digit log-probs, F_0 a population CDF fitted on training respondents, $\bar{Q}$ its mean CDF on a calibration pool, Π_{CDF} the exact L_2 projection onto monotone CDFs, and α chosen on calibration respondents. This is deliberately the most favourable treatment: it hands the simulator a correctly located population distribution for free and asks only whether its *person-specific* deviations are worth anything.

What the baseline is allowed. D_m is a ridge over the one-hot observed answers onto the threshold indicators, projected onto monotone CDFs. Its penalty is retuned at *every* budget m , on the same calibration respondents that select α . This is not a detail: holding the penalty fixed overfits the small budgets, and in development that alone manufactured an apparent 15% relative gain for the simulator at $m=100$ which vanished once the baseline was tuned at each budget. A comparison across label budgets that does not retune baseline capacity is measuring the baseline’s misspecification.

Model	raw S	α^*	m^* (of 60)
Qwen3.6-35B-A3B	0.172	0.85	3.07
Qwen2.5-32B	0.213	0.30	2.15
Qwen2.5-72B	0.202	0.45	1.66
Qwen3.8-27B	0.183	1.10	1.31
QwQ-32B-Preview	0.202	0.70	1.29
granite-4.2-30b	0.243	0.45	0.45
gemma-4-12B-it	0.267	0.35	0.37
Mistral-Small-24B-2501	0.324	1.00	0.33
<i>5 further models, App. U</i>			
granite-4.1-8b	0.356	0.20	0.02
Apertus-8B-2509	0.286	0.00	0.00

Table 5: Answer-equivalent on IPIP-NEO-120, averaged over both panels. S is the normalised ordinal RPS of the raw readout. α^* is the selected calibration coefficient. Raw accuracy does not order the ladder; α^* does.

Protocol. Conditioning and target items are disjoint by construction; respondents are split into a training block, a calibration half on which every hyperparameter is selected, and an evaluation half on which every reported number is computed frozen; uncertainty is a paired bootstrap over evaluation respondents. Appendix S gives the full specification, including the reverse-orientation correction, the common-support rule and the multiplicity control.

11 What a persona is worth

We price 89 cells: every model in our artifact set, on IPIP-NEO-120, SD3, IPIP-FFM-50 and HEXACO, on two disjoint respondent panels, plus the IPIP-NEO-300 unseen-item holdout of §11.1.

The instrument is stable. Across the 32 (model, instrument) pairs scored on both panels, m^* on the `val` panel and on the `test2` panel correlate at Spearman $\rho=0.78$ ($p=1.7 \times 10^{-7}$). The price is a property of the model and instrument, not of a panel draw. Without this, nothing below would be worth reporting.

The price. Table 5 gives the IPIP-NEO-120 ladder. Median m^* per inventory is 0.00 on HEXACO, 0.06 on IPIP-FFM-50, 0.32 on IPIP-NEO-120, 0.40 on SD3 and 0.61 on the IPIP-NEO-300 holdout (App. U). The strongest persona we measure on the short form is worth 3.07 of 60 answers; the weakest is worth exactly zero, in the strong sense that calibration selects $\alpha=0$ and discards its person-specific deviations entirely. Across all 89 cells the median is 0.19 answers.

Reading the number. 3.07 of 60 is a real, replicable, non-zero price, and it is small. Both halves matter. A practitioner with a 60-item instrument and a budget for three more questions should ask them. A practitioner choosing between open models should read the ladder, because the spread across models — 0 to 3.07 — is larger than any effect reported for prompt optimisation on these systems.

11.1 Transfer to an unseen item family

The splits above put every target facet’s siblings in the conditioning half. The IPIP-NEO-300 holdout removes that crutch: 300 respondents answered the full inventory, 120 of whose items are exactly the short form used everywhere else, and the other 180 — spanning all 30 facets — are items the same person answered that no arm has been fitted on, with only 150 training respondents behind them. This is the thinnest supervision we measure and the most favourable regime for a simulator, and it is where the price is highest: 5.38 of 120 answers, against 3.24 of 60 on the short form, with the model ordering preserved. A persona is worth most exactly where human labels are scarcest.

Presentation matters as much as quantity. Given its persona as m literal (item, answer) pairs rather than a 30-line profile, the same model prices at 0.00, 0.19, 0.49, 1.10 for $m=5, 20, 60, 120$ — monotone, as it must be — while the profile built from those same 120 answers is worth almost five times the transcript of all of them (α^* 0.9 against 0.3). The direction is not universal: Apertus-0.5B agrees (0.13 against 0.06) but gemma-4-E2B reverses it (0.22 against 0.54). Three models cannot settle which presentation wins; what the price shows is that the choice is worth several observations and belongs in the report (App. V).

12 Why some personas are worth more

The ladder is not ordered by accuracy. The raw score of the best-priced model (0.172) and of gpt-oss-20b (0.188) are close, but their prices differ by more than a factor of ten. Across all 89 cells, raw score correlates with m^* at $\rho = -0.48$, and the *calibrated* score — the thing one would naively report — does not correlate with it at all.

What does order the ladder is α^* , the share of a model’s person-specific deviation that survives calibration: $\rho = +0.74$ ($p < 10^{-13}$), and $\rho = +0.72$ after partialling out raw score. A model whose

Corpus	cells	LLM	prior	decoder	$\bar{w}^*$	sig. cells	b
HEXACO	40	0.222	0.159	0.119	0.040	4/80	+0
IPIP-FFM-50	24	0.239	0.166	0.121	0.000	0/48	+0
IPIP-NEO-120	54	0.247	0.156	0.110	0.001	2/108	+0
SD3	42	0.215	0.167	0.130	0.000	2/84	+0

Table 6: Incremental value of a persona given the respondent’s own observed answers. $\bar{w}^*$ is the mean selected convex weight on the persona. Significant cells are *uncorrected* counts, shown to make the multiplicity visible: across all 426 one-sided tests, one survives Benjamini–Hochberg at $q < 0.05$, and it reverses sign on its own disjoint panel.

forecast barely moves between people offers calibration nothing to keep; Apertus-8B has $\alpha^* = 0$ and a price of exactly zero, despite a middling raw score.

We state the limit of this plainly. α^* is fitted against the same score m^* is derived from, on a disjoint respondent half but the same objective, so it is a tight proxy rather than an independent cause. We therefore also measured between-person dispersion directly from the saved belief tensors, with no fitting and no labels, and we report the cluster-level variance-ratio statistics of §14 against m^* as well. Those cluster-level statistics do *not* predict the price (all $|\rho| \leq 0.17$, none significant over the 42 matched cells) — which is itself informative: a population-level dispersion audit does not tell you what a persona is worth to an individual.

13 The boundary: personas are redundant given real answers

The price is quoted against a model given m answers. The complementary question is whether a persona adds anything to a model that already has *all* of the respondent’s observed answers. It does not.

We screen 160 cells. For each, three combination families are fitted on the calibration half and frozen on the evaluation half: a convex weight between the conditional model and the calibrated persona; a linear combiner taking the persona’s CDF as additional features beside the conditional model’s; and a gradient-boosted combiner over the same features. The control in each case is the identical combiner, of identical capacity, fitted on identical rows, without the persona features — so the comparison isolates the persona and nothing else.

The selected convex weight on the persona is 0 in 127 of 160 cells, 0.05 in the remaining 33, and

never higher. The linear combiner finds a nominally significant positive increment in 0 of 160 cells. Across the full grid — 426 one-sided tests — 35 are nominally significant at 5%, which is what chance produces, and exactly *one* survives Benjamini–Hochberg at $q < 0.05$: Qwen3.8-27B on HEXACO, `val` panel, gradient-boosted combiner, $+0.00092$ [$+0.00046, +0.00140$], $q=0.026$.

That cell does not survive scrutiny: on the disjoint `test2` panel the gain reverses sign (-0.00034), and at four times the panel size it falls to $+0.00006$ — a small-sample artefact, as is every other nominally significant HEXACO cell, all of which sit on the same 64-respondent panel. We report it rather than filter it out; a boundary claim that discards its one exception is not a boundary claim.

We also report the null as a null rather than as a failed superiority test: the comparison is pre-specified, the combiner is given every advantage its control is given, and the selected weight — not merely the p -value — is what concedes. A method whose own validation selection sets the persona’s weight to zero in 127 of 160 cells is making a stronger statement than an inconclusive test would.

This is the sharp statement the price makes precise. A persona carries a small, real, model-scaling amount of personal information — and a handful of the person’s actual answers renders it redundant.

Beyond one questionnaire. The structural defect is not a property of IPIP-NEO-120. On the Short Dark Triad (17,740 respondents, 27 items) the same inequality holds on 27/27 items, with a penalty of 8.67 points against IPIP’s 7.75, and the constant again outscores a real respondent (0.744 vs 0.664) in 200 of 200 bootstraps. §11 prices two further inventories, IPIP-FFM-50 and HEXACO, on the same footing.

14 Where the variance is lost

The dispersion shortfall that §8 measures has a locatable cause, and it matters here because §12 shows a persona’s price is governed by exactly this quantity. Reading each model’s own belief from its digit logprobs, before any sampling, separates three stages: what the model believes, what decoding emits, and what parsing recovers. The shortfall is already present in the belief — a low-dispersion model’s forecast barely moves between respondents, so there is nothing for decoding to discard and nothing for temperature to reveal. Rais-

ing T adds dispersion without adding alignment. All mass lands on valid answer tokens throughout, so accounts blaming mode collapse on difficulty *emitting* numbers are unsupported. Appendix O gives the full ladder, the per-model tables and the reachability analysis.

Notably, the cluster-level statistics this analysis produces do *not* predict the individual-level price (§12). A population dispersion audit and an individual answer-equivalent are different measurements, and the paper needs both.

15 Related work

Personality conditioning of LLMs is validated by administering psychometric inventories to the conditioned model (Serapio-García et al., 2023; Jiang et al., 2024), with well-documented prompt sensitivity (Han et al., 2026; Gupta et al., 2024); simulation from transcripts achieves high fidelity (Park et al., 2024) at a per-person cost that motivates the cluster-level shortcut, and evolutionary prompt search (Guo et al., 2024; Prasad et al., 2023; Zhou et al., 2022) supplies the optimiser. None of this reports what a persona is worth. Classifier two-sample tests are standard for comparing distributions (Lopez-Paz and Oquab, 2017), but the absolute-error objective these systems optimise is minimised by a conditional central tendency, which is why we anchor the price to a proper score instead.

Closest to us are methods that *fix* simulator distributions rather than price them: distributional fine-tuning on survey data (Suh et al., 2025), two-stage shift alignment (Huang et al., 2025), mean-preserving dispersion adjustment over value profiles (Abels et al., 2026) — the family our α belongs to — and three-axis fidelity from small pilots (Choi et al., 2026). That calibration and distributional fine-tuning are established is exactly why we claim neither; we contribute the unit their gains should be quoted in. Simulation grounded in self-reports (Park et al., 2026) and behaviour-trained foundation models (Huang et al., 2026; Binz et al., 2025) target richer behaviour than questionnaire response, and the answer-equivalent applies to them unchanged wherever a held-out response distribution exists.

16 Conclusion

We priced respondent simulators in the currency a practitioner spends: questions asked of a real per-

son. The answer-equivalent is stable across disjoint panels, orders models consistently, and is governed by how much person-specific deviation survives calibration rather than by accuracy. The strongest open persona we tested is worth 5.38 of 120 answers on an unseen item family and the weakest exactly zero, and across 160 cells a persona adds nothing to a model that already holds the respondent’s own answers. The instrument is what we ask the field to adopt: it makes “our simulator is better” a claim with units.

Limitations

Our audit covers one corpus (IPIP-NEO-120), one clustering pipeline, one optimisation family, and five models; the belief-level measurement of §14 covers eleven open-weight models spanning 7B–235B, six families and three generations, and the cross-instrument replication carries the structural findings and the identity result but not the full suite, which needs sampled simulator outputs where SD3 has only the two belief readouts of App. B; belief-level measurement anywhere needs logprob access the API models in the audit do not provide, and three current large mixture-of-experts models — GLM-5.3-Flash, Qwen3.8-Flash-Next and DeepSeek-V4-Flash — could not be read at all on this protocol, returning NaN or zero mass on the answer tokens or shipping no chat template, so the candidate pools here reach 35B and not beyond; individual readouts are reproducible bit-for-bit across clusters and software stacks but *not* across accelerator architectures, where 7.7% of modal answers flip, so the per-item beliefs we release should be read as machine-specific even though the aggregates they feed move by at most 0.016 (App. F); the within-family trend rests on three points and the cross-family comparison on eleven, so both are directional rather than fitted laws; all models are read with thinking disabled and the assistant turn prefilled, and the four least confident of them (belief entropy above 1 nat) should not have their above-human total dispersion read as calibrated uncertainty, though low confidence cannot inflate VR_{between} , which is computed from persona means; two elicitation details proved to matter more than we expected — reading these models through a chat-completions server returned almost no mass on the answer tokens where the forward pass recovers 0.99–1.00, and harmony-format models require the assistant turn to open a channel before the pre-

fill, without which mass on the scale falls to 0.39 — so a null result in this literature may report a read-out rather than a model; precision varies across the models we report — nine run at bf16, gpt-oss-20b ships MXFP4 and is dequantised to bf16 at load, and Qwen3-235B is an FP8 checkpoint served at FP8 — and a further FP8 checkpoint resisted measurement through both available routes, so we make no claim that individuation is invariant to quantisation; the proposition is general to the absolute-error metric in (1), but the empirical magnitudes are specific to this setting. The model answers we re-score were produced by prior runs we did not control, at temperatures we did not set, and one released run was committed with entirely missing answers and is excluded. Our human ceiling is a single same-cluster respondent, which underestimates what a well-specified individual-level twin could achieve; we make no claim about individual-level methods. C2ST-AUC depends on classifier capacity and on $n=40$ per side, so its absolute value should be read as a lower bound on separability, and §Q shows most of what it reads is response style; we fold the AUC as $\max(\text{AUC}, 1 - \text{AUC})$, which places the null above 0.5 (0.517 signed against 0.569 folded) and makes our small- n floor a property of this classifier rather than of C2ST; and DF treats items independently and so ignores within-respondent answer dependence. The corpus is self-reported, skewed toward respondents aged 18–25 and toward the USA, UK and Canada, so neither the clusters nor the ceilings should be read as representative of any broader population.

Limits specific to the price. (i) m^* is a threshold crossing on a curve estimated with noise; we report the interpolated value and a paired bootstrap on the loss gap, but a single cell’s crossing is fragile — the top cell’s gap excludes zero on one panel and not on the other, which is why we lead with the ladder and the cross-panel rank correlation rather than with any one number. (ii) α^* is fitted against the same score m^* is derived from, on a disjoint respondent half but the same objective, so it is a tight proxy for retained person-specific deviation rather than an independent cause of it. (iii) The price is quoted against the specific conditional model we use; a stronger conditional model would lower every m^* , so our numbers are upper bounds on what a persona is worth. (iv) We price questionnaire response, not behaviour. The four corpora are four disjoint populations with no shared respondents, so we cannot demonstrate that one

person’s identity transfers across behavioural domains, and we make no such claim. (v) The data contain one response per person–item cell and no repeated administration, so we cannot estimate the irreducible response noise that bounds any simulator, and m^* therefore cannot be expressed as a share of attainable performance. (vi) Our null results are reported as nulls under a pre-specified comparison with Benjamini–Hochberg control across the 426-test grid, but we did not pre-register an equivalence margin, so “adds nothing” should be read as “adds nothing detectable at this sample size and effect scale”.

Finally, we show that current personas are worth little on this instrument; we do not show that no method can be worth more, and the instrument is designed to register it when one is.

Ethics statement

We use a public, de-identified corpus of self-reported personality questionnaire responses and add no new human-subjects data collection. Our results are cautionary: they indicate that current cluster-level personality simulations are readily distinguishable from real respondents and under-represent human response variance, and so should not be used as substitutes for human participants in research, product, or policy decisions. Reporting an evaluation number without its constant floor risks overstating the fidelity of synthetic respondents, which is the specific harm this paper aims to reduce. Simulating demographic or personality groups also risks reinforcing stereotypes when cluster descriptions are treated as characterisations of real people rather than as statistical summaries.

A Reproducibility

Every analysis in this paper is deterministic given its inputs and requires no LLM calls except the belief-level measurement of §14. The released corpus and committed run artefacts reconstruct the scoring, selection and orientation analyses. Some inputs are too large for the repository and are not in it: the refreshed belief tensors behind App. L, the per-respondent score arrays behind App. M, the $n=512$ forecast tensors behind App. D, and the cross-cluster and cross-architecture replication readouts behind App. F. These accompany the submission as one archive (568 files, 92 MB compressed) carrying a per-file SHA-256 manifest and the CPU commands that regenerate each number

above from it; no GPU and no model weights are required, because the archive holds the saved model outputs rather than the runs that produced them. We verified that path end to end: from the archive alone, the intervals of App. L, the frequencies of App. M, the eight paired differences of App. D and the deviations of App. F all reproduce as printed. We do not claim the list is exhaustive; where an analysis consumes a tensor, the tensor is in the repository only if this appendix says so, and otherwise in the archive. We ran the full analysis pipeline independently on two clusters with different operating systems, Python versions and library versions (scikit-learn 1.2 vs 1.7, pandas 2.1 vs 3.0). Every reported figure we compared — the λ^* values, the orientation deltas, the variance components, and the null-calibration ARIs of App. J — agreed to the digits we print, except the $k=4$ reproduction ARI, which is 0.946 on one environment and 0.951 on the other. Code and per-cell results are released. The belief simplices (40 personas $\times$ 120 items $\times$ 5) per cluster are included for the eleven models of §14, with the persona ordering fixed by the released corpus. Runs that failed the validity guard ship too, marked invalid, so the record shows what was measured and rejected. One transform maps model answers onto the corpus’s recoded scale and is imported by every analysis that compares the two.

B A disjoint conditioning/target replication

Every belief result in the body conditions on scores computed from the same items it is then scored on, so a sceptic can say the models reconstruct arithmetic rather than predict a person. We remove that by construction. Items are split within each scale, the persona is built only from the input half, and the readout covers only the target half: two input and two target items per IPIP facet (60/60), four input and five target per SD3 trait (12/15). No cluster label is used — the released IPIP clustering is derived from all 120 items and would reimport target information — and the prior and the decoder are fitted on training respondents whose target answers are never seen. The five score bins are fixed equal-width bands of the 0–100 range, not fitted. This design differs from the body’s in more than the split: the persona is one generated line per facet rather than the body’s per-cluster prose, respondents are sampled globally rather than within clusters, and only half the items are scored. It is a replication

under a stricter design, not a controlled ablation of overlap.

On 256 held-out respondents the second fact of §14 is unqualified and the first is weakened. Every model still loses to the empirical prior, by 0.098–0.204 across all four model–instrument cells. The identity effect survives in three of those four: on SD3 by +0.010 [+0.006, +0.016] for Qwen2.5-7B and +0.017 [+0.013, +0.020] for Granite-4.1, and on IPIP by +0.004 [+0.002, +0.005] for Qwen2.5-7B — but Granite-4.1 on IPIP gives +0.002 [−0.001, +0.005], which does not exclude zero.

One thing follows and one does not. Positive identity gains occur with conditioning and target items disjoint, in three of four cells, so item overlap is not necessary for them. What this design cannot do is apportion the body’s IPIP gain between overlap and identity, because it changes the persona text, the respondent sample and the scored items as well as the split: Qwen2.5-7B’s 0.006 there and 0.004 here are not two measurements of one quantity, and an interval excluding zero in one design beside one that does not in another is no evidence of a decrease. What survives every cell is that the simulator loses to a prior that ignores the persona.

C Does the replacement change a decision?

An audit that only condemns a metric leaves the practical question open: if you adopt the replacement, do you ship something different, and is it better? We fix one candidate set and select twice.

Candidates are the eleven belief models of §14, re-run under the disjoint-item prompts of App. B; the body’s readouts cannot serve, since they condition on scores computed from the answers being scored. Qwen3-235B is excluded: its FP8 checkpoint read correctly on one allocation and returned near-zero mass on the answer tokens on others, so it fails the same ≥ 0.5 mass criterion we apply throughout, and reading it through a different execution path would give one candidate an elicitation route the other ten did not have. Both rules see the same probability forecasts, never greedy answers: rule A ranks by expected answer similarity, the audited S_0 ; rule B by $S_{1/2}$. Selection uses 128 respondents; evaluation uses 128 more, disjoint from those, from the 20,000 training respondents on which both baselines are fitted, and from App. B’s panel. Only the selected models are

run on the evaluation panel.

The rules disagree. Rule A selects Qwen2.5-72B ($S_0 = 0.753$); rule B selects Qwen3.5-9B ($L_{1/2} = 0.187$) — models an order of magnitude apart in size. On the untouched panel rule B’s pick is better by 0.010 [0.004, 0.016] under $L_{1/2}$ (lower is better), paired over respondents. The task is ordinal prediction, so that is the outcome we report; categorical log-loss agrees and is secondary, at 1.37 nats [1.26, 1.48]. The log-loss gap is the larger number but the weaker argument, since it mostly registers the 72B’s overconfidence rather than its ordinal accuracy — its errors are confident, not distant. Its one advantage is independence: neither selection rule uses log-loss, whereas $S_{1/2}$ is rule B’s own criterion, evaluated on disjoint respondents.

Two caveats belong with this. It is one candidate set, one split and one instrument, so it shows the evaluation changed a decision for the better — not that it reliably does. And the improvement is relative: both picks lose to the training-fitted empirical prior (0.155, 1.41 nats), so the replacement recommends a smaller model and simultaneously reports that no candidate beats ignoring the persona. Adding the degenerate candidates makes the point sharper still. With the constant vector in the pool, rule A selects *it* over every model ($S_0 = 0.766$ against the 72B’s 0.753), which is Prop. 1 arriving as a deployment recommendation; rule B selects the prior. The constant’s log-loss is infinite, and we report it as such rather than clipping it.

D Which scoring rule selects best?

App. C compares two rules on one instrument, and says plainly that this cannot show the replacement *reliably* chooses better. Here we widen it: nine scoring rules, twelve open-weight models spanning 0.5B to 35B across six families, and four instruments, two of which the body never uses.

The rules span three classes. Strictly proper: $S_{1/2}$ (ranked probability score), Brier, log-loss, spherical. Weakly proper: RMSE of the forecast mean, and mode accuracy. Improper: the audited S_0 , MAE of the forecast mean, and Wasserstein distance to the observed point mass, which coincides with S_0 by construction here and so is not independent evidence. Instruments are IPIP-NEO-120, SD3 and IPIP-FFM-50 on five points, and HEXACO-PI-R on *seven* — the last is the only place the pipeline is exercised at $K \neq 5$. The

first three are keyed from published item keys; the HEXACO distribution we use ships none, so its reverse-keyed items are inferred from each item’s correlation with the rest of its scale, and HEXACO results carry that caveat. Each rule selects on 512 respondents and is graded on 512 disjoint others, with baselines fitted on a further 8,000; all 96 read-outs put at least 0.92 of their mass on the answer tokens.

We score a selector by the *regret* its pick incurs, normalised within the candidate set so 0 is the best of these twelve models and 1 the worst, and averaged over the strictly proper evaluators *excluding the selector’s own rule*, so no rule is graded on the criterion it optimised. Because eight rules are compared against one reference, we report Holm-adjusted p and Bonferroni simultaneous intervals, and call a difference resolved only when both agree. The four differences below with intervals excluding zero all carry Holm $p \leq 0.01$; the three strictly proper alternatives carry $p = 1.00$.

rule	class	regret vs. $S_{1/2}$ (Bonferroni CI)
RMSE of mean	weak	+0.000 [+0.000, +0.000]
Brier	strict	+0.003 [−0.008, +0.025]
log-loss	strict	+0.004 [−0.007, +0.022]
spherical	strict	+0.007 [−0.002, +0.029]
MAE of mean	improper	+0.025 [+0.007, +0.055]
mode accuracy	weak	+0.238 [+0.084, +0.262]
S_0 audited	improper	+0.298 [+0.250, +0.368]
Wasserstein	improper	+0.298 [+0.250, +0.368]

Each row grades the two selectors on the same evaluators — the strictly proper rules minus both contestants’ own criteria — so a rule is never compared on a set containing its opponent’s criterion but not its own. Differences are bootstrap means over respondents, $B=4000$; intervals are Bonferroni simultaneous bounds holding jointly over all eight comparisons. Every improper rule selects worse than $S_{1/2}$, and the audited S_0 is the worst of them. The gap is not marginal: mean regret is 0.259 for S_0 against 0.003 for $S_{1/2}$. On SD3, S_0 selects a model whose regret is 0.98 — all but the worst of the twelve — while every strictly proper rule selects the best. An evaluator using the audited metric on this pool would ship very nearly the worst simulator available.

The number depends on the pool. With eight closely matched instruction-tuned models the same comparison gives +0.028 [+0.011, +0.059]: still resolved, ten times smaller. Regret is normalised within the candidate set, so it measures position within that pool and cannot be read across pools. What it does show is where in *this* pool the audited

metric lands: near the bottom on SD3, where its selection carries regret 0.98, while the proper rules select the best.

Three things this does not say. The strictly proper rules remain mutually indistinguishable, so the evidence is for the class boundary, not a ranking inside it. RMSE of the forecast mean — only weakly proper — is not distinguishable from $S_{1/2}$ at all: it selects the same model on every instrument, and its paired difference is exactly zero. Eliciting the mean is sufficient on this candidate set, and we have no basis for preferring a strict rule over it. And regret is normalised within the candidate set, so 0 means the best of these twelve, never good: every candidate still loses to the persona-free prior of App. B.

Mode accuracy is the other collapse, at +0.238. Discarding the distribution and keeping the argmax is as damaging as the metric this paper audits. If a practitioner takes one operational rule from this work, it is that a simulator must be scored on the distribution it induces, not on the answer it would most likely give.

Sample size decided the weaker version of this. At 128 respondents per panel with eight candidates the S_0 comparison was +0.070 [−0.017, +0.273] — unresolved. Quadrupling the panels resolved it; widening the pool made it large.

E When does impropriety actually bite?

App. D shows S_0 selecting all but the worst simulator from a heterogeneous pool. It would be convenient to conclude that optimising S_0 is therefore harmful wherever it appears. We tested that directly and it is false.

The system this paper audits was built by evolving cluster-level system prompts to maximise S_0 . We take those evolved genotypes and the base genotypes the search started from, assemble both with the original code, and read the same belief distribution under each — scoring with the objective that was optimised and with two proper rules. The prediction we registered before running was that S_0 would improve while the proper scores would not.

model	ΔS_0	$\Delta L_{1/2}$	$\Delta \log\text{-loss}$
Qwen3.8-27B	+0.060	−0.039	−0.235
granite-4.2-8B	+0.046	−0.046	−0.630
gpt-oss-20b	+0.052	−0.043	−0.251
gemma-4-12B-it	+0.070	−0.068	−0.619

Evolved minus base, 256 held-out respondents over four clusters, paired bootstrap. Every interval

excludes zero and *every* direction is an improvement: $L_{1/2}$ and log-loss are losses, so the negative signs are gains. The prediction is falsified on all four models. Across respondents the change in S_0 and the change in $S_{1/2}$ correlate at $r = 0.96\text{--}0.99$, and 85–97% of respondents improve on both at once.

The two results are not in tension. S_0 is maximised by a constant vector (Prop. 1) — a *point mass*, not a flat distribution — so its optimum is under-dispersed. Writing $\bar{\sigma}$ for a model’s mean predictive standard deviation, the Spearman correlation between $S_{1/2}$ as a reward and $\bar{\sigma}$ is positive on all four instruments (+0.71 to +0.86): the proper score consistently prefers the more dispersed candidate. S_0 does not, and its correlation varies — +0.03 on IPIP, −0.09 on IPIP-FFM, −0.40 on HEXACO and −0.97 on SD3 — with the failure tracking the penalty. On SD3, where the correlation is −0.97, S_0 ships a model with $\bar{\sigma} = 0.48$ against the proper rule’s 1.10 and incurs regret 0.98; on IPIP and IPIP-FFM, where it is near zero, both rules select the same model and the regret is zero. These are correlations over twelve candidates per instrument and we read them as descriptive, not as an identified mechanism.

What we can say about optimisation is narrower. Evolutionary prompt search improved both S_0 and the proper scores on all four models tested, so optimising an improper objective did not degrade the artefact here. We have since measured the dispersion of the candidates a prompt search can visit (App. M): over twelve persona conditions the rules disagree on two models of four, and on one of those the disagreement is stable under resampling, with S_0 taking the less dispersed candidate. The degenerate direction is therefore reachable by prompt search in this candidate set, at a small observed cost. Model selection over a heterogeneous pool is the other case we measured: an over-confident weak model is already most of the way to a point mass, so S_0 rewards it and ships it.

The pattern across the two settings is narrower than “never optimise S_0 ”. In the candidate sets we measured, S_0 disagreed with a proper rule far more visibly in the model-selection pool, where it shipped a near-worst simulator, than in the prompt pool, where the selected candidates scored within 0.001 CRPS of each other. We do not convert that into a claim that one setting is more dangerous than the other: the two are not measured on a common scale (App. M). We report this because it bounds

our own claim: the audit establishes that S_0 is improper and that it selects badly, not that every system built with it is worse for having used it.

F Is a belief readout a measurement?

Every number in App. D treats a readout as a stable property of a model. That is an assumption, and this paper’s own Limitations record the readout *path* moving mass on the answer tokens from 0.39 to 0.99, so we tested it rather than appealing to determinism. Two models and two instruments were re-read with the same script, seed and respondent panels on independent systems.

Same architecture and dtype, different everything else. Between two clusters running different software — torch 2.11.0 with transformers 5.17.0 against torch 2.13.0 with transformers 5.16.1, both on H200 and both on the `bf16` path — the readouts are *bit-identical*: maximum deviation 0 over the full belief simplex and 0 of 76,800 item–respondent pairs changing their modal answer. This is a statement about two small models on one numeric path, and §F.1 below shows it does not extend to a change of dtype.

Different architecture. Against A100 (`sm_80`) instead of H200 (`sm_90`), on the same `bf16` path, they are not. Mean absolute deviation per probability is 0.008–0.029, and 5,946 of 76,800 modal answers flip — 7.7%. The divergence is not a mismatched comparison: the four cells score identical respondents, and the flips sit precisely on near-ties. Where a modal answer flips, the median gap between the model’s top two levels is 0.051 against 0.507 where it does not; for one cell that median gap is 0.000, exact ties resolved differently by different kernels.

What survives. The scores largely do. Averaged over the four cells, the change is 1.6×10^{-3} for S_0 , 3.4×10^{-3} for mode accuracy and 4.7×10^{-3} for $S_{1/2}$; the largest single-cell change is 1.6×10^{-2} , for $S_{1/2}$ on gemma-4-E2B/SD3. We do not convert these into a ratio against the regret differences of App. D: raw score changes and candidate-range-normalised regrets are different units and the comparison would be meaningless. What we can say is bounded and direct — a change of architecture moves a model’s score by at most 0.016 here, while it moves 7.7% of that model’s individual answers.

We had expected the distribution-based scores to be the *stable* ones, which would have given §8

a second argument independent of propriety. They are the least stable of the three, and we report that rather than the argument we went looking for. The practical reading is narrower: a per-item LLM “belief” is not reproducible across accelerators and should not be published as though it were, while an average over hundreds of items and respondents is stable enough to carry an effect an order of magnitude larger than the noise.

One arm is missing from *this* comparison, and it is not a minor one. V100 (sm_70) would have tested the fp16 path; the build available when we ran these checks ships no Volta kernels, so the comparison above spans two Hopper-class and one Ampere-class device only. The fp16 path is nevertheless the path on which part of §14’s table was measured, which we address next.

F.1 The table is not on a single numeric path

Auditing the run records for §14 showed that its rows were not all produced on one device class. h4_variance_ladder_hf.py selects bf16 at sm_80 and above and fp16 below it, and four rows — Mistral-Small-24B, Qwen2.5-7B, Qwen2.5-32B and QwQ-32B — were read on V100 (sm_70) in fp16, while the remainder were read on H200 (sm_90) in bf16. The appendix above tests bf16 against bf16; it never tested the path those four rows used.

We therefore re-read all four on sm_90 in bf16, holding the model, the script, the panel (40 personas per cluster) and the respondents fixed, so that only the device class and dtype differ. The belief readout is deterministic given a model and a panel — the seed enters only the sampled answers — so these are not repeated draws:

Model	sm_70 fp16	sm_90 bf16	Δ
Qwen2.5-7B	0.430	0.427	−0.003
Qwen2.5-32B	0.343	0.340	−0.003
QwQ-32B	0.753	0.824	+0.070
Mistral-Small-24B	0.579	0.771	+0.192

Two of the four reproduce to 0.003 in VR_{total} . Two do not: QwQ moves 9% and Mistral 33%, in both cases because the model emits flatter answer distributions on the newer path — Mistral’s mean belief entropy rises from 0.729 to 0.984 — which inflates the within-persona term. Panel size is not the explanation: on sm_90 Mistral moves 0.002 between 40 and 160 personas.

The individuation column is not affected. The quantity §14 actually rests on is $VR_{between}$, and it survives the change of path where the total does not.

Re-reading ten of the eleven rows on sm_90 bf16 at a common 160-persona panel (QwQ at 40, the largest panel on which it returns a readout) reproduces every published $VR_{between}$ to within 0.0064, and the four fp16-origin rows are no worse than the six bf16-origin ones:

Model	pub.	re-read	Δ
Qwen2.5-7B [†]	0.193	0.187	−0.006
Qwen2.5-32B [†]	0.220	0.221	+0.001
Qwen2.5-72B	0.295	0.292	−0.003
Mistral-24B [†]	0.177	0.176	−0.001
QwQ-32B [†]	0.154	0.148	−0.006
gpt-oss-20b	0.126	0.123	−0.003
Granite-4.1-8B	0.207	0.203	−0.004
Apertus-8B	0.130	0.125	−0.005
GLM-4.7-Flash	0.265	0.263	−0.002
Qwen3.5-9B	0.167	0.163	−0.004

[†] published row measured on sm_70 fp16.

Mistral is the clearest case: its VR_{total} moves 0.192 between paths while its $VR_{between}$ moves 0.001. That is what the mechanism predicts. A change of numeric path alters how sharp each persona’s answer distribution is, which is a within-persona quantity; it barely moves where those personas’ *means* sit, which is what $VR_{between}$ measures.

The practical consequence is narrower than the mixed path first suggests. The individuation claims of §14 and the ordering of models by $VR_{between}$ stand: every published value reproduces within 0.0064 on one path. What does not transfer is the total-dispersion column and the within/between share derived from it, for the two models whose distributions sharpen. We leave the published table as measured and flag the affected statistic rather than restating numbers, and note that App. L is uniformly sm_90 bf16 throughout.

One further caution for anyone reusing the script: at 160 personas the QwQ read returned NaN for every cell while exiting normally, and was caught only by the validity guard that rejects an all-missing run. The same read at 40 personas succeeded. We have not traced the cause; a partially rather than wholly degenerate version of that fault would not trip the guard.

G Split sensitivity of the baselines

Re-drawing which 100 respondents are sampled at a fixed 60/40 ratio moves the parameter-free cluster-mean baseline by 0.54–0.77 points within a cluster over 20 draws, and the four-cluster average by 0.23 (pooling clusters gives 1.33, mixing cluster differences into split variation). Prior work’s gain is +0.72 points on one fixed split. This is context

for how far a split can move a score, not a test of that gain: variation in baseline *levels* does not bound a *paired* improvement.

H What is a persona worth, in respondents?

“It loses to a constant” invites the reply that the constant is a strawman. So we restate the finding in a unit that is harder to dismiss: how many real respondents you would have to sample before a plug-in estimate of the cluster’s response distribution matches what the persona gives you.

Drawing m real respondents, forming the per-item empirical distribution, and sampling a synthetic population from it yields distributional fidelity 0.723 at $m=1$, 0.797 at $m=2$, 0.915 at $m=21$ and 0.923 at $m=34$. The curve is evaluated at the same sizes as the audited scores — forty synthetic against forty held-out humans — because DF depends on sample size, so a curve built at any other size cannot calibrate them. Reading the audited systems off it:

System	DF	Resp.-equiv.
Evolutionary LLM persona	0.783	2
Constant cluster mean	0.795	2
Human ceiling	0.939	off-curve

An evolutionary-optimised persona is worth about **two** real respondents: two is the smallest evaluated m whose mean DF exceeds the persona’s 0.783, and the constant sits at the same place. This is a threshold on the mean curve, not an equivalence test. The human ceiling of 0.939 is not crossed anywhere on the evaluated grid, up to $m=500$ (0.938), so we do not convert it into respondent units — a statement about the grid we searched, not a proof that no m reaches it. This is the same result as §4 in different units, and it is the version we would put to a practitioner weighing synthetic respondents against a small pilot survey.

I The C2ST feature ladder

A metric that separates simulators from humans at AUC 0.994 invites an obvious objection: perhaps the classifier is detecting formatting rather than anything about personality. We test our own metric with a feature ladder, each rung scored identically against the same anchors. Orientation is handled per feature family: style features describe how a respondent uses the scale, so both sides sit on the literal scale the item was asked on, while content

features compare trait-oriented quantities. The full-item rung reproduces the suite’s 0.994.

Features	LLM	Human	Const.
Full 120 items	0.994	0.564	0.926
5 style scalars only	0.894	0.584	0.988
Items, ipsatised	0.989	0.566	0.986
30 facet scores only	0.991	0.565	0.908
Forward-keyed only	0.990	0.568	0.810
Reverse-keyed only	0.993	0.583	0.843

Five content-free scalars — per-respondent mean and standard deviation, extreme -response rate, longest identical run, and forward-versus-reverse consistency — attain 80% of the full above-chance AUC on their own. The objection has real force, and we state it plainly: **response style alone reproduces most of the separability**, which is a statement about what suffices, not a decomposition of what the full classifier uses.

It is not only style. Ipsatising each respondent (removing their own mean and standard deviation, i.e. the two largest style features) still leaves AUC 0.989, and facet scores alone give 0.991. Style and content are redundant routes to the same conclusion rather than one masking the other. The human ceiling is also stable at 0.56–0.58 across every rung, so the anchor does not move with the feature set.

J Cluster validity in full

Since stability-based model selection is common, the general point is worth stating: at large n , bootstrap agreement must be read against a covariance-matched null or it certifies nothing.

No *separation* criterion supports $k=4$. Silhouette falls overall from 0.134 at $k=2$ to 0.053 at $k=15$, with minor upward steps, taking 0.082 at $k=4$; Davies–Bouldin is also minimised at $k=2$; the gap statistic increases monotonically and selects no interior k . A silhouette below 0.1 indicates essentially no separated structure.

Stability looks at first like a defence: bootstrap ARI over 40 resamples is 0.977, 0.952, 0.932, 0.903 at $k=2, 3, 4, 5$ and falls away after (0.797, 0.581), so $k=4$ looks close to the largest reproducible k .

That reading does not survive a null calibration. Rebuilding the pipeline on structureless surrogates — a Gaussian matched to the real 30×30 facet covariance, and a copula matched to the covariance and every per-facet marginal — reproduces the stability number exactly: ARI 0.911 and 0.921 at silhouette 0.075 and 0.078, against the real data’s 0.910 at 0.082. A separated 4-component control

confirms the test has power (silhouette 0.672, ARI 1.000).

At $n=410,168$ any smooth unimodal cloud with this covariance is partitioned reproducibly, so ARI ≈ 0.91 measures estimator variance, not multimodality, and neither criterion supports $k=4$. The clusters are not demographic artefacts either (Cramér’s V of 0.055–0.057 against sex, age band and country); they are a reproducible discretisation of a continuum, which bounds the cluster-conditional signal any persona can exploit. Since stability-based model selection is common, the general point is worth stating: at large n , bootstrap agreement must be read against a covariance-matched null or it certifies nothing.

K What the persona channel carries

A persona reaches the model only through `get_modifier_bisect`, which writes each score as one of five words on width-20 bins; each cluster’s genotype names five traits and five of the thirty facets, and every other sentence of the system prompt is shared. Ten words therefore carry the respondent’s identity, and two respondents sharing those ten bins would receive byte-identical prompts and so must receive identical beliefs.

We estimate what that channel permits. Within each cluster we fit $\mathbb{E}[X_j \mid \text{code}]$ on respondents *excluding* the forty personas, then evaluate on those forty, so the quantity is directly comparable to $\text{VR}_{\text{between}}$ and is never fitted on its own evaluation set.

Conditioning available to the model	Dec. SD
Ten scores, five words (<i>audited system</i>)	0.482
same, with boosted trees	0.483
Ten scores, written as numbers	0.512
All 35 scores, five words each	0.707
All 35 scores, written as numbers	0.726
<i>best of the eleven</i>	0.295

Three things follow. The quantisation is not the binding constraint: writing numbers instead of words buys 0.03. The channel is not degenerate: all forty personas in all four clusters receive distinct codes, so no model is forced to answer two personas identically. And a boosted-tree decoder, which can use interactions, lands in the same place (0.483 against 0.482); that is consistent with the additive fit being close to the conditional mean, though two similar SD ratios do not prove it.

The oracle is fitted on this population and so knows its code-to-response mapping, which a zero-shot model does not; we therefore do not claim

that a model *ought* to reach 0.482. The comparison rules out one explanation only — that the conditioning carries too little information to individuate — and that is the explanation the flatness of $\text{VR}_{\text{between}}$ across eleven models most invites.

Uncertainty in §14. Identity gain and the prior comparison use 2000 percentile-bootstrap draws over respondents, 40 per cluster with equal cluster weight. The permutation control is not sampled but computed exactly, as the expectation of every donor row against every respondent; the prior and decoder are fitted once on the excluded respondents and held fixed, so intervals cover respondent sampling, not fitting.

Uncertainty on A. Persona rows are resampled jointly across the model, the decoder and the actual respondents, so paired comparisons stay paired; the decoder fit is held fixed, so the intervals describe persona-sampling uncertainty rather than fit uncertainty. Over 2000 draws the human reference is 0.498 [0.461, 0.517] and the eleven models span 0.033 [0.028, 0.036] for Apertus-8B to 0.183 [0.168, 0.193] for Qwen2.5-72B. The intervals are informative enough to separate models that $\text{VR}_{\text{between}}$ ranks together: Qwen2.5-32B and Granite-4.1 differ by 0.013 in spread but their amplitudes do not overlap, [0.105, 0.122] against [0.071, 0.089]. Comparisons are made on the paired draws, not by asking whether marginal intervals overlap — two overlapping marginals can still have a difference interval excluding zero, as GLM-4.7 and Qwen3-235B do (+0.017 [+0.009, +0.025]) — and Qwen2.5-32B exceeds Granite-4.1 by +0.034 [+0.028, +0.040] despite a 0.013 gap in spread.

Where the shortfall falls. Each cluster names five facets, so twenty items have their own facet stated and one hundred are covered only at trait level. The named items carry $1.51\times$ more predictable-from-code signal (an SD ratio) (reference 0.671 against 0.444), and 10 of 11 models individuate them more in absolute terms, Qwen3-235B being the exception (0.224 named against 0.264 elsewhere). Measured against those references the ratio is *lower* where the prompt is most specific — 0.353 on the named items against 0.433 on the rest, in the same direction for 10 of 11 models — but we report this descriptively and draw no causal conclusion from it, for two reasons. A facet score is computed from its own four items,

so $\mathbb{E}[X_j \mid \text{facet}]$ is close to an arithmetic identity that a model without the scoring key cannot invert, while traits have the same structure diluted over twenty-four items; the named-item reference is inflated by more than the trait-level one, which alone could produce the gap. And the eleven models share personas, references and prompt structure, several within one family, so they are not independent replicates and we do not attach a p -value to the count.

Widening the channel does not fix it. The obvious remedy is to send more: we re-ran three models with every other word of the prompt held fixed and the ten modifier phrases replaced by the numbers themselves. The effect is inconsistent in magnitude — $\text{VR}_{\text{between}}$ rises for Granite-4.1 ($0.207 \rightarrow 0.246$) and gpt-oss ($0.126 \rightarrow 0.146$) but falls for Apertus ($0.130 \rightarrow 0.096$) — and, using the same joint draws so the comparison is paired, every model moves resolvably: A shifts by $+0.053 [+0.045, +0.059]$ for Granite-4.1 and $+0.008 [+0.005, +0.012]$ for gpt-oss but $-0.013 [-0.016, -0.009]$ for Apertus. The effect is therefore not noise; it is real and model-dependent in sign. It is also inconsistent in direction, the aligned amplitude moving $0.081 \rightarrow 0.134$ for Granite-4.1 and $0.039 \rightarrow 0.047$ for gpt-oss but $0.033 \rightarrow 0.020$ for Apertus. Granite gains on both counts — its correlation with the oracle rises from 0.285 to 0.453 — so for that model the richer encoding is a real improvement, while Apertus loses spread and direction together. On three models a richer encoding is therefore a plausible but unreliable remedy, and we report it as a pilot rather than a recommendation.

The alignment does not depend on the oracle. Because r is measured against a fitted target, an oracle missing structure the models track would understate it. It does not. Boosted trees on the same code change r by at most 0.012 and a ridge on the continuous scores by at most 0.034; an oracle given all thirty-five scores lowers r substantially, by up to 0.168, which is the expected direction since models only ever see the ten-score code. The ordering of models is unchanged throughout. Two independent computations agree where theory demands it: the ceiling as a standard-deviation ratio is 0.482 and the oracle’s correlation with actual respondents is 0.491, which coincide for a conditional mean. The check that assumes no oracle at all — correlating each model’s persona means directly against the forty respondents’ own answers — gives 0.076–

0.248 against the oracle’s 0.491, so models recover between a sixth and a half of the attainable person-level alignment.

L Current-generation models, with intervals

The eleven-model table in §14 predates three of the four families below; gpt-oss-20b is re-measured rather than new. We repeated the belief-level measurement on one current model from each, under the identical protocol (40 personas per cluster, four clusters, exact readout from the retained probability vectors), and added the interval the original table lacks. Some readouts are missing for Gemma-4 — 30 persona–item readouts of 19,200, one in 640, all in cluster 1 where they are 30 of 4,800 (coverage 0.99375); those cells are held out of the moments rather than summed to zero, which would enter as an impossible answer of 0 and inflate the between-persona term. Every other model here has complete coverage. Resampling is over personas — the exchangeable unit for a between-persona variance — with 2000 draws; because the protocol assigns the same personas to every model, each replicate applies one set of persona indices to all four, so the differences below are paired.

Model	Params	VR_{with}	VR_{betw} [95% CI]	VR_{tot}
Granite-4.2-8B	8B	0.729	0.155 [0.143, 0.160]	0.757
Gemma-4-12B-it	12B	0.336	0.334 [0.303, 0.345]	0.506
gpt-oss-20b (MoE)	21B	1.120	0.127 [0.118, 0.130]	1.130
Qwen3.8-27B	27B	0.869	0.219 [0.197, 0.229]	0.911
human	—	—	1.000	1.000

The measurement replicates. gpt-oss-20b appears in both tables from independent runs. The new run returns $\text{VR}_{\text{between}} = 0.127$ against 0.126, $\text{VR}_{\text{within}} = 1.120$ against 1.121, and $\text{VR}_{\text{total}} = 1.130$ against 1.131. We report this because it is our only same-model repeat under this variance protocol (App. F tests readout stability separately), and it bears on §14 rather than on the new models.

The 0.30 bound is not a law. Gemma-4-12B measures $\text{VR}_{\text{between}} = 0.334 [0.303, 0.345]$, above the ≈ 0.30 figure that held across all eleven earlier models — but the interval’s lower end sits essentially *at* that figure, so we read this as evidence that the ceiling is a property of the models we happened to measure rather than as a decisive refutation of it. The qualitative claim is unchanged: 0.334 is still about a third of the human value. We have scoped §14 accordingly rather than restating the bound as universal.

The two failure modes are separable. Gemma-4 and gpt-oss-20b fail in opposite directions, and the intervals separate them. gpt-oss-20b carries *above*-human total dispersion ($VR_{\text{total}} = 1.130$) of which almost none individuates: the between-persona share of belief variance is 0.016 [0.014, 0.018]. Gemma-4 is its mirror — a between-persona share of 0.504 [0.466, 0.524] but collapsed in magnitude ($VR_{\text{total}} = 0.506$). It is not alone in that: Qwen2.5-32B and Qwen2.5-72B in the eleven-model set have between-persona shares of 0.485 and 0.429. The paired difference, gpt-oss minus Gemma-4, is -0.488 [$-0.507, -0.451$]. Near-human total spread is therefore not evidence of individuation, and individuation is not evidence of human-like spread; a single variance ratio cannot distinguish the two, which is the argument of §14 restated on models chosen after it was written.

What does not separate. Of the six paired comparisons of the between-persona share, five exclude zero and one does not: Granite-4.2 against Qwen3.8 is -0.008 [$-0.017, +0.003$]. Their point estimates, 0.068 and 0.076, must not be read as an ordering. We note it because the same two models *do* separate on VR_{between} itself (-0.064 [$-0.073, -0.049$]): which quantity is reported decides whether these two models are distinguishable, and the point estimates alone would not have revealed that.

M Is the degenerate direction reachable by prompt search?

App. E reports that prompt search improved S_0 and the proper scores together, and notes that this alone cannot establish whether the degenerate optimum is out of reach — we had not measured the dispersion of the candidates a search can actually visit. This appendix measures it.

Design. We cross six persona directives with the base and evolved genotypes, giving twelve candidates scored on the same 192 respondents. Three directives are intended to push towards concentration (*decisive*: avoid middle-of-the-scale answers; *extreme*: answer at the ends; *consistent*: similar items receive the same number), one is intended to push towards spread (*varied*), one asks for the spread of a single real respondent (*calibrated*), and one is the unmodified prompt. Each candidate is scored under S_0 and under CRPS, and we record its mean predictive SD. The verdict is fixed in code

before the run: REACHABLE iff the candidate S_0 selects is strictly less dispersed than the one the proper rule selects.

Stability. A strict inequality between two close candidates is not a result, so we resample respondents (2000 draws, paired — all conditions are scored on the same respondents) and *re-run the selection inside each replicate*, so the reported stability includes the selection uncertainty rather than freezing the point-estimate picks.

Model	S_0 selects	rules differ	REACH.
Granite-4.2-8B	evolved/varied	0%	0.0%
Gemma-4-12B-it	evolved/varied	0%	0.0%
Qwen3.8-27B	evolved/varied	77%	77.4%
gpt-oss-20b	evolved/calibrated	99%	99.2%

Two point-estimate disagreements; one of them stable. On Granite-4.2 and Gemma-4 both rules select the same candidate in every one of the 2000 replicates: nothing to separate. On Qwen3.8 the point estimate fires REACHABLE, but the proper rule’s own choice is unstable — it takes *calibrated* in 77% of replicates and *varied* in the rest — so we report it as inconclusive rather than as a second instance. On gpt-oss-20b the disagreement is stable: S_0 selects *evolved/calibrated* in 100% of replicates, CRPS selects *evolved/varied* in 99%, the two rules differ in 99.2%, and S_0 ’s candidate is less dispersed by 0.0453 [0.0439, 0.0465]. That is the phenomenon Prop. 1 predicts, produced by an ordinary prompt search rather than by a constructed counterexample.

What these frequencies are, and are not. The 90% stability line that separates “stable” from “inconclusive” above is a descriptive cut we chose, not an inferential guarantee. It is not load-bearing: gpt-oss alone qualifies at 80, 90, 95 and 99%, and neither disagreement qualifies at 99.5%. A bootstrap selection frequency is the stability of a choice under resampling, not the probability that a claim is true. The resampling is stratified to match the design — 48 respondents drawn within each of four clusters, as H21 draws them — but re-selecting inside each replicate measures selection instability on one evaluation panel, which is not an independent deployment test. We therefore read App. M as an exploratory finite-candidate result.

The cost of following it is small. Magnitude matters for the practical claim. On gpt-oss-20b the contested pair is *evolved/varied* ($S_0 = 0.7173$, CRPS = 0.1667, $\bar{\sigma} = 0.899$) against *evolved/calibrated*

($S_0 = 0.7234$, $\text{CRPS} = 0.1677$, $\bar{\sigma} = 0.854$). Following S_0 costs 0.0010 of CRPS — about 0.6% of the better candidate’s score — and buys a 5% reduction in dispersion. So the degenerate direction is reachable here, but the observed cost of taking it is small. We deliberately do not compare that figure to the 0.98 regret reported for model selection in App. E: that regret is normalised within a candidate set and averaged over evaluators, and this is a raw CRPS difference, so the two are not on one scale. What the two experiments jointly show is that the disagreement occurs in both settings, not that one is quantifiably more dangerous than the other.

The directives that ask for confidence do not deliver it. Across all four models neither *decisive* nor *extreme* is selected by either rule, and both lower S_0 relative to the unmodified prompt. The reason is not that they concentrate in the wrong place: measured against the matching base prompt they *raise* mean predictive SD in 13 of 16 model–genotype comparisons. Their names describe an intention, and the measured effect usually runs the other way, so we report them as instruments that failed rather than as evidence about where S_0 ’s optimum lies. *Calibrated*, which S_0 selects on gpt-oss, asks for the spread of a single real respondent rather than for confidence. It is not the only condition that lowered dispersion — *evolved/varied* sits below *evolved/base* on all four models, and both rules select it on Granite-4.2 and Gemma-4. A search that reliably produced low-dispersion candidates might well find more disagreement than we do; our directive set is a weak probe of that space.

N Finite-sample behaviour of the anchors

Both anchors are estimated from $n=40$ per side, which for a classifier over 120 features is a small-sample regime. We sweep n on human-only data, where the true answers are known: two disjoint samples from the same cluster must be indistinguishable, so the honest C2ST value is exactly 0.5 and any excess is estimator bias.

n	20	40	80	160	320	640
C2ST human	—	0.574	0.542	0.536	0.527	0.517
DF human	0.915	0.939	0.957	0.969	0.978	0.985
C2ST const.	—	0.914	1.000	1.000	1.000	1.000

Two corrections follow, in opposite directions, and both land on their theoretical values. Fitting $y = y_\infty + cn^{-1/2}$ over the non-degenerate range gives an extrapolated C2ST human ceiling of **0.500** and an extrapolated DF human ceiling

of **1.000** — exactly what two samples from the same distribution must score. So the C2ST ceiling of 0.574 at $n=40$ is *entirely* finite-sample classifier bias and should be read with its n attached, while the DF ceiling of 0.939 is *entirely* plug-in Wasserstein bias and is pessimistic. Correcting the latter widens the gap to the simulator’s 0.783 rather than narrowing it.

Below $n \approx 40$ *this* classifier stops separating even the constant from real humans (the dashed entries above are the degenerate 0.5). That floor is a property of the configuration we used, not of C2ST: relaxing the minimum leaf size from 20 to 5 restores separation at $n=20$. We therefore recommend reporting every anchor together with the n and the classifier settings it was measured at, and treating any small- n floor as something to establish per configuration rather than to inherit from us.

O Where the variance is lost, in full

Knowing the simulator is mis-dispersed does not say *where* the dispersion goes. We read the model’s own belief before any sampling occurs, on open-weight models served locally.

Two stages. For each persona and item we take the model’s next-token distribution over the answer tokens $\{1, \dots, 5\}$ directly from the logprobs, then sample each item from its own belief and score the resulting population. One implementation detail matters: hybrid reasoning models emit a `<think>` token first — with probability ≈ 1 — so a naive readout sees no mass on the scale at all; we disable thinking and prefill the assistant turn.

Result. What is collapsed is the *individuating* part of the belief, not the belief’s total spread. In the 2024–25 models both are small; in the 2026 reasoning-first models the total spread reaches or exceeds the human marginal while individuation stays as low as ever.

The statistic matters. The quantity comparable to a human *across-respondent* SD is not the average within-persona conditional SD but the SD of the persona-*mixture* distribution, from the law of total variance $\text{Var}(X_j) = \mathbb{E}_i[\text{Var}(X_j | i)] + \text{Var}_i(\mathbb{E}[X_j | i])$, computed exactly from the retained probability vectors. The within-persona term alone would understate a faithful simulator, which can hold sharp per-persona beliefs and still match human variance through differing persona

means.

Model	Params	VR _{with}	VR _{betw}	VR _{tot}
Qwen2.5-7B-Instruct	7B	0.367	0.193	0.430
Qwen2.5-32B-Instruct	32B	0.242	0.220	0.343
Qwen2.5-72B-Instruct	72B	0.367	0.295	0.488
Mistral-Small-24B	24B	0.538	0.177	0.579
QwQ-32B (reasoning)	32B	0.729	0.154	0.753
Qwen3-235B-A22B-2507	235B	0.695	0.258	0.758
Qwen3.5-9B	9B	1.058	0.167	1.078
GLM-4.7-Flash (MoE)	29B	1.209	0.265	1.242
gpt-oss-20b (MoE)	21B	1.121	0.126	1.131
Granite-4.1-8B	8B	0.386	0.207	0.457
Apertus-8B-Instruct	8B	0.927	0.130	0.939
<i>human</i>	—	—	—	1.000

Essentially all probability mass lands on the five valid answer tokens (0.97–1.00), so no model has difficulty *emitting* a number. Belief entropy nonetheless spans 0.42–1.38 nats (maximum $\ln 5 = 1.609$), and we can offer no clean account of it: it tracks neither release date (Granite-4.1, 0.49) nor reasoning-tuning (QwQ 0.97; the standard-instruct Apertus 1.24). The 235B model is the one the audited system used, so this measures that system, not a proxy.

Total dispersion and individuation are decoupled. The decomposition says something the total does not, and the evidence for this is existential rather than correlational. gpt-oss-20b and Qwen2.5-72B invert cleanly: the former has $2.3\times$ the total dispersion of the latter (1.131 vs 0.488) and *less than half* its individuation (0.126 vs 0.295). The rank correlation over the eleven models is $\rho = -0.30$ with a bootstrap 95% interval of $[-0.92, 0.42]$.

QwQ-32B sharpens it: its dispersion is almost all per-item uncertainty (0.729 of 0.753), and it individuates less (0.154) than far smaller models.

The Qwen2.5 family isolates this at fixed family and varying scale: individuation rises monotonically, $0.193 \rightarrow 0.220 \rightarrow 0.295$ at 7B, 32B, 72B, while the total does not, moving $0.430 \rightarrow 0.343 \rightarrow 0.488$.

One bound survives across the whole set — a $34\times$ parameter range, six families, three generations, dense and mixture-of-experts, and both standard and reasoning-tuned models: no model here exceeds $\text{VR}_{\text{between}} \approx 0.30$ — a bound on this set, not a law: Gemma-4-12B, measured later, reaches 0.334 (App. L). Individuation stays under half the human value throughout, and neither the largest model nor the most dispersed one is the best at it.

The persona helps, and still loses. Two questions have run together: does the simulator reproduce the cluster’s distribution, and does it identify

this respondent? Scoring each persona against that person’s own answers under normalised CRPS, and permuting belief rows among respondents within a cluster — which leaves every population statistic identical while destroying identity — separates them. All eleven score better on the correct assignment than the permuted one by 0.004–0.017, every paired interval excluding zero, yet all remain worse than a cluster empirical prior fitted on *other* respondents — which uses cluster membership but no persona — by 0.036–0.095. With conditioning and target items disjoint, the gain is resolved in three of four tested cells and unresolved for Granite-4.1 on IPIP, while all four still lose to the prior (App. B).

The two facts reconcile exactly, and without any fitted reference. Writing I for the identity gain above, $\bar{q}$ for the model’s within-cluster mean CDF and D_Q for the across-persona variance of its CDFs,

$$R(P_0) - R(Q) = I - D_Q - [R(\bar{q}) - R(P_0)], \quad (3)$$

an identity in measured quantities, exact on our data to machine precision. It locates the loss. The bracket — how much worse the model’s *mixture* is than the prior — is 0.038–0.090, whereas the net identity contribution $D_Q - I$ lies within ± 0.005 for every model, in absolute value below a tenth of that model’s mixture excess. Spread across personas can help or hurt slightly; what decides the comparison is that the mixture itself is wrong.

The prompt is not what caps individuation. A persona is individuated only through a five-word quantisation of ten numbers, but that channel carries more than the models extract: a supervised conditional-mean decoder fitted on held-out respondents reaches 0.482 on these same personas, where the best model delivers 0.295, and no two personas share a code (App. W). The decoder is a reference, not a bound — a mis-calibrated simulator could exceed it by dispersing wrongly — so it shows the conditioning carries signal.

How much of it points the right way. $\text{VR}_{\text{between}}$ measures how far apart a simulator places its persona means, not whether that spread identifies the right respondent. Correlating each model’s persona means against the oracle’s over the forty personas — both on the corpus’s recoded scale — gives $r = 0.16$ –0.52. The aligned amplitude $A = \frac{1}{J} \sum_j r_j \text{sd}(M_j) / \text{sd}(H_j)$, in units of human item SD, is 0.033–0.183, against 0.498 [0.461, 0.517]

for real respondents: the best reaches about a third of that, the weakest under a tenth. It separates models VR_{between} ranks together — GLM-4.7 and Qwen3-235B sit 0.007 apart in spread, yet differ in A by $+0.017$ [$+0.009, +0.025$] — and rises with scale across Qwen2.5 in three non-overlapping steps ($0.067 \rightarrow 0.116 \rightarrow 0.183$).

A simulator can therefore look more dispersed simply by being less confident: VR_{total} confounds uncertainty with individuation, and VR_{between} is what a persona claim should be judged on.

Total dispersion is not uniformly low: it is 0.758 for the largest 2024–25 model and exceeds the human marginal for three others (1.078, 1.131, 1.242). Nor is it lost at decoding: sampling retains almost all of it ($1.078 \rightarrow 1.061$, $1.243 \rightarrow 1.218$, $1.131 \rightarrow 1.115$). What stays limited at every scale is *individuation*.

For the low-dispersion models the spread is likewise not discarded at decoding; it was never present, so recovering it there means *adding* dispersion, not revealing it: $p^{1/T}$ lifts Qwen2.5-7B from 0.43 to 1.02 at $T=5$ while supplying none of the alignment A measures. And since all mass lands on valid answer tokens, accounts blaming mode collapse on difficulty *emitting* numbers are unsupported here.

P Is $k=4$ a partition at all?

The audited systems condition on a $k=4$ k -means partition of the 30 facet scores. We reproduce it ($ARI = 0.946$) and test $k \in \{2, \dots, 15\}$ under three preprocessings: no criterion supports $k=4$, and the bootstrap stability that appears to defend it is matched exactly by structureless surrogates (App. J). The clusters are a convenient partition, not a discovered structure — which is why our results are stated for the partition the systems use rather than for personality types.

Q What is C2ST actually reading?

A metric that separates simulators from humans at AUC 0.994 invites the objection that the classifier detects formatting rather than personality. A feature ladder (App. I) shows the objection has real force: five content-free response-style scalars already recover most of the separability above chance. It is not only style — ipsatising each respondent, or using facet scores alone, still separates nearly as well — so style and content are redundant routes to the same conclusion. We therefore rec-

ommend reporting C2ST with its style-only rung alongside it, so readers can see how much of any separability is carried by response style. A simulator that closed only the style gap would look far better on the headline number while remaining just as distinguishable on content.

The anchors are estimated at $n=40$ per side. Appendix N sweeps n and shows the C2ST human ceiling is entirely finite-sample classifier bias (extrapolating to 0.500) while the DF ceiling is entirely plug-in bias (extrapolating to 1.000, i.e. pessimistic). Below $n \approx 40$ our classifier configuration stops separating at all, so we report every anchor with the n and the settings it was measured at rather than proposing a universal floor.

R Cluster conditioning, per cell

No persona study we know of reports the wrong-persona control, so no reported persona effect is currently attributable to the persona. We score every simulator against every cluster’s humans — 80 cells — and report diagonal dominance descriptively.

Metric	Matched	Mismatched	Δ
S	0.728	0.663	+0.065
DF	0.782	0.705	+0.077
C2ST	0.996	0.985	−0.010
VR	0.431	0.436	−0.005

Matched comparisons improve S and DF; they show no improvement in C2ST or VR. We report these gaps descriptively and attach no p -value: a permutation that shuffles the 80 cell labels freely destroys the matched structure of the five 4×4 matrices, and the models share clusters and personas, so its null is not the right one. Rescoring fixed outputs against different human populations also does not isolate a causal effect of changing the prompt. This is the behavioural counterpart of §8 — the persona moves which items get endorsed, and not who is endorsing them — and it makes the wrong-persona control as necessary a baseline as the constant floor.

S Answer-equivalent protocol in full

Conditioning and target items are disjoint by construction, split within each scale, so the simulator is never scored on items its persona was built from. Respondents are split three ways: a training block on which F_0 , the prior and every D_m are fitted; a calibration half of the scored panel on which α , the ridge penalty and any combination weight are

selected; and an evaluation half on which every number we report is computed, frozen. Uncertainty is a paired bootstrap over evaluation respondents. Where the corpus stores reverse-recoded answers — IPIP-NEO-120 does, on 43 of 60 targets; SD3 does not — the model’s simplex is reversed on those items before scoring. Omitting that step is invisible in every variance statistic and wrong in every score.

T The orientation audit in full

Before comparing anything, the two sides must be on the same scale. They are not.

The released corpus stores human answers *reverse-recoded* so that a higher number always means more of the trait, while the simulator is shown the literal item text — “Cheat to get ahead” — and asked how accurately it describes it, i.e. it answers *raw*. On the 55 negatively-keyed items of the IPIP-NEO-120 the two scales run in opposite directions, and every published model-vs-human comparison is computed across that flip.

Two signatures make this unambiguous. Negatively-keyed items average 3.373 against 3.425 for positively-keyed ones — level, whereas raw self-reports on undesirable content sit far lower. And “Cheat to get ahead” carries a human mean of 4.15 out of 5.

The effect is large and it runs *against* the simulator. Correcting the orientation raises the correlation between the model’s and the humans’ per-item mean profiles from +0.09 — indistinguishable from no relationship at all — to +0.63, and raises S from 0.660 to 0.739.

Three conclusions survive, with their invariances:

- **Proposition 1 is invariant.** MAD and GMD are computed on the human distribution alone, and both are reflection-invariant.
- **The variance ratio is exactly invariant,** since $\text{sd}(6 - x) = \text{sd}(x)$. The under-dispersion finding of §8 is untouched.
- **The constant still wins every cell.** Reorientation closes the gap from 13.6 to 5.8 points, but the constant cluster mean beats the corrected model in 20 of 20 cells.

Correcting the defect in fact *sharpens* the paper’s claim. The corrected model scores 0.739, *above* the 0.721 of a real member of the target cluster — while a classifier separates that same

model from real members at AUC 0.994 (§8). A metric that ranks a near-perfectly detectable simulator above the humans it fails to imitate is not measuring human-likeness under any reading.

We recommend an orientation check as a standing first step for benchmarks of this kind: correlate the simulator’s per-item mean profile against the population’s, and treat a near-zero correlation as a scale defect until shown otherwise.

U Answer-equivalent, every cell

Table 8 prices every (model, instrument, panel) cell.

Corpus	cells	raw S	calibrated S	median m^*	max m^*
HEXACO	16	0.223	0.158	0.00	0.17
IPIP-FFM-50	16	0.236	0.165	0.06	0.99
IPIP-NEO-120	30	0.246	0.154	0.32	3.24
IPIP-NEO-300	8	0.162	0.092	0.61	5.38
SD3	18	0.216	0.161	0.40	1.77

Table 7: Answer-equivalent by inventory.

V The unseen-item-family holdout

Everything above splits items *within* each scale, so every target facet is also represented in the conditioning half. The IPIP-NEO-300 holdout removes that: 300 respondents answered the full inventory, 120 of whose items are exactly the short form used everywhere else, and the remaining 180 — spanning all 30 facets — are items the same person answered that no arm has been fitted on. With only 150 training respondents, per-item supervision is two orders of magnitude thinner than on IPIP-NEO-120, which is the regime most favourable to a simulator.

It is indeed the most favourable regime we measure, and by a clear margin: the best persona is worth **5.38** of 120 answers, against 3.24 of 60 on the short form, and the ordering of models is preserved. A persona is worth most exactly where human labels are scarcest — which is the regime a practitioner is in when simulation is tempting at all.

Profile beats transcript. The same model, on the same items, was also given its persona as m literal (item, answer) pairs instead of a 30-line trait profile:

m^* rises monotonically with the transcript length, as it must — 0.00, 0.19, 0.49, 1.10 at

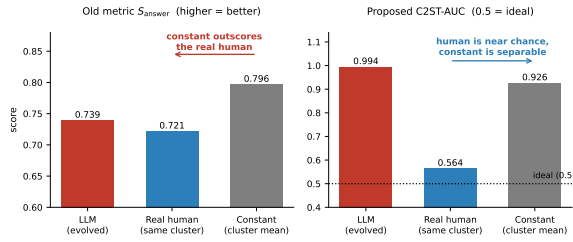


Figure 2: Corrected orientation, 20 cells. Left: under the old metric the constant outscores a real human. Right: under C2ST-AUC (lower is better) the human sits near chance (0.5) while model and constant are both highly separable. Only the right panel places the human above the constant.

$m=5, 20, 60, 120$ — but the 30-line profile computed from those same 120 answers is worth **5.38**, almost five times the raw transcript of all of them, and retains $\alpha^*=0.9$ against the transcript’s 0.3. The aggregation is doing the work, not the model’s reading of individual responses. How the observations are presented matters more than how many there are.

W The human ceiling, per cell

The decisive check is what a *real* member of the target cluster scores. A cluster-level digital twin cannot do better than being an actual member of the cluster; that is the definition of success.

A randomly drawn same-cluster respondent scores 0.721 — and is beaten by the constant cluster mean in 20 of 20 cells (Table 3, Figure 2, left). This is exactly Proposition 1 realised: a real human *is* a draw from P_j , so their expected score is $1 - \text{GMD}/4$, which the constant’s $1 - \text{MAD}/4$ must exceed. The predicted values (0.7996 constant, 0.7221 faithful) match the measured ones (0.796, 0.721) to within 0.4 points.

A measure that ranks a fixed vector above a genuine group member is not measuring group membership. Any ranking it induces over simulators — including the ranking prior work uses to select prompts — is therefore uninterpretable as evidence about personality fidelity.

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Model	Corpus	LLM raw	+calib.	α^*	m^*	m_{interp}^*
Apertus-8B-2509	HEXACO	0.2092	0.1553	0.2	1.0	0.02
gemma-4-12B-it	HEXACO	0.2804	0.1554	0.0	0.0	0.00
granite-4.2-30b	HEXACO	0.2273	0.1554	0.0	0.0	0.00
granite-4.2-8b	HEXACO	0.2666	0.1554	0.0	0.0	0.00
gpt-oss-20b	HEXACO	0.1895	0.1554	0.0	0.0	0.00
GLM-4.7-Flash	HEXACO	0.1969	0.1554	0.0	0.0	0.00
Qwen3.6-35B-A3B	HEXACO	0.2052	0.1554	0.0	0.0	0.00
Qwen3.8-27B	HEXACO	0.1927	0.1554	0.0	0.0	0.00
Apertus-8B-2509	HEXACO	0.2227	0.1599	0.2	1.0	0.17
gemma-4-12B-it	HEXACO	0.2783	0.1602	0.0	0.0	0.00
granite-4.2-30b	HEXACO	0.2311	0.1602	0.0	0.0	0.00
granite-4.2-8b	HEXACO	0.2721	0.1602	0.0	0.0	0.00
gpt-oss-20b	HEXACO	0.1987	0.1602	0.0	0.0	0.00
GLM-4.7-Flash	HEXACO	0.2042	0.1602	0.0	0.0	0.00
Qwen3.6-35B-A3B	HEXACO	0.2043	0.1602	0.0	0.0	0.00
Qwen3.8-27B	HEXACO	0.1932	0.1602	0.0	0.0	0.00
Qwen3.6-35B-A3B	IPIP-FFM-50	0.1770	0.1607	0.5	1.0	0.51
Qwen3.6-35B-A3B	IPIP-FFM-50	0.1772	0.1620	0.6	1.0	0.99
granite-4.2-30b	IPIP-FFM-50	0.2697	0.1635	0.6	1.0	0.05
gpt-oss-20b	IPIP-FFM-50	0.1823	0.1636	1.2	1.0	0.03
Apertus-8B-2509	IPIP-FFM-50	0.2148	0.1637	0.3	1.0	0.01
GLM-4.7-Flash	IPIP-FFM-50	0.2284	0.1639	0.3	0.0	0.00
granite-4.2-8b	IPIP-FFM-50	0.3470	0.1643	0.4	0.0	0.00
granite-4.2-30b	IPIP-FFM-50	0.2667	0.1643	0.5	1.0	0.64
Qwen3.8-27B	IPIP-FFM-50	0.1998	0.1643	0.7	0.0	0.00
gemma-4-12B-it	IPIP-FFM-50	0.2773	0.1647	0.1	0.0	0.00
gpt-oss-20b	IPIP-FFM-50	0.1825	0.1650	1.0	1.0	0.47
Qwen3.8-27B	IPIP-FFM-50	0.1996	0.1652	0.8	1.0	0.47
GLM-4.7-Flash	IPIP-FFM-50	0.2242	0.1662	0.3	1.0	0.29
Apertus-8B-2509	IPIP-FFM-50	0.2207	0.1668	0.3	1.0	0.16
granite-4.2-8b	IPIP-FFM-50	0.3447	0.1673	0.7	1.0	0.08
gemma-4-12B-it	IPIP-FFM-50	0.2701	0.1677	0.0	0.0	0.00
Qwen3.6-35B-A3B	IPIP-NEO-120	0.1703	0.1429	0.8	5.0	3.24
Qwen3.8-27B	IPIP-NEO-120	0.1826	0.1481	1.0	2.0	1.50
Qwen3.6-35B-A3B	IPIP-NEO-120	0.1728	0.1496	0.9	3.0	2.90
Qwen2.5-72B	IPIP-NEO-120	0.1981	0.1501	0.5	2.0	1.28
granite-4.2-30b	IPIP-NEO-120	0.2432	0.1511	0.4	1.0	0.49
gemma-4-12B-it	IPIP-NEO-120	0.2693	0.1515	0.3	1.0	0.28
gpt-oss-20b	IPIP-NEO-120	0.1833	0.1518	0.5	1.0	0.18
Qwen3.8-27B	IPIP-NEO-120	0.1843	0.1522	1.2	2.0	1.12
GLM-4.7-Flash	IPIP-NEO-120	0.2382	0.1523	0.1	1.0	0.04
granite-4.2-8b	IPIP-NEO-120	0.3701	0.1524	0.4	1.0	0.01
Apertus-8B-2509	IPIP-NEO-120	0.2756	0.1524	0.0	0.0	0.00
Qwen2.5-72B	IPIP-NEO-120	0.2063	0.1533	0.4	3.0	2.04
Qwen2.5-7B	IPIP-NEO-120	0.2481	0.1534	0.1	1.0	0.12
Qwen2.5-32B	IPIP-NEO-120	0.2134	0.1536	0.3	3.0	2.15
granite-4.1-8b	IPIP-NEO-120	0.3555	0.1536	0.1	1.0	0.04
Qwen3.5-9B	IPIP-NEO-120	0.1859	0.1540	0.6	1.0	0.37
QwQ-32B-Preview	IPIP-NEO-120	0.2016	0.1542	0.7	2.0	1.29
granite-4.2-30b	IPIP-NEO-120	0.2426	0.1560	0.5	1.0	0.40
gemma-4-12B-it	IPIP-NEO-120	0.2648	0.1569	0.4	1.0	0.46
granite-4.2-8b	IPIP-NEO-120	0.3823	0.1575	0.5	1.0	0.39
Mistral-Small-24B-2501	IPIP-NEO-120	0.3239	0.1576	1.0	1.0	0.33
gpt-oss-20b	IPIP-NEO-120	0.1935	0.1577	0.9	1.0	0.25
GLM-4.7-Flash	IPIP-NEO-120	0.2468	0.1577	0.3	1.0	0.31
gpt-oss-20b	IPIP-NEO-120	0.1931	0.1580	0.6	1.0	0.41
Qwen3.5-9B	IPIP-NEO-120	0.1896	0.1583	0.4	1.0	0.26
Apertus-8B-2509	IPIP-NEO-120	0.2971	0.1585	0.0	0.0	0.00
GLM-4.7-Flash	IPIP-NEO-120	0.2472	0.1588	0.3	1.0	0.10
Qwen2.5-7B	IPIP-NEO-120	0.2551	0.1589	0.1	1.0	0.08
granite-4.1-8b	IPIP-NEO-120	0.3560	0.1591	0.3	1.0	0.00
Apertus-8B-2509	IPIP-NEO-120	0.2855	0.1591	0.0	0.0	0.00
Qwen3.6-35B-A3B	IPIP-NEO-300	0.1172	0.0860	0.9	8.0	5.38
Qwen3.8-27B	IPIP-NEO-300	0.1473	0.0906	1.1	3.0	2.23
Qwen3.6-35B-A3B	IPIP-NEO-300	0.1462	0.0927	0.3	2.0	1.10
granite-4.2-30b	IPIP-NEO-300	0.2261	0.0930	0.7	1.0	0.73
Qwen3.6-35B-A3B	IPIP-NEO-300	0.1478	0.0935	0.2	1.0	0.49
Qwen3.6-35B-A3B	IPIP-NEO-300	0.1470	0.0938	0.2	1.0	0.19
Qwen2.5-7B	IPIP-NEO-300	0.2166	0.0944	0.3	1.0	0.02
Qwen3.6-35B-A3B	IPIP-NEO-300	0.1484	0.0944	0.1	1.0	0.00
Qwen3.8-27B	SD3	0.1861	0.1536	1.6	2.0	1.77
Apertus-8B-2509	SD3	0.1866	0.1553	1.4	2.0	1.51
GLM-4.7-Flash	SD3	0.1893	0.1564	1.1	2.0	1.19
granite-4.2-8b	SD3	0.2220	0.1596	1.1	1.0	0.85
Qwen3.6-35B-A3B	SD3	0.1956	0.1599	1.2	1.0	0.78
granite-4.2-8b	SD3	0.2288	0.1606	1.2	1.0	0.56
Apertus-8B-2509	SD3	0.1968	0.1614	1.6	1.0	0.48
granite-4.2-30b	SD3	0.2309	0.1616	0.8	1.0	0.72
GLM-4.7-Flash	SD3	0.1930	0.1617	1.4	1.0	0.34
granite-4.2-30b	SD3	0.2352	0.1618	1.0	1.0	0.37
Qwen3.8-27B	SD3	0.1936	0.1622	1.9	1.0	0.27
Qwen3.6-35B-A3B	SD3	0.2027	0.1623	1.6	1.0	0.22
gpt-oss-20b	SD3	0.1843	0.1628	1.8	1.0	0.11
granite-4.1-8b	SD3	0.2660	0.1628	0.6	1.0	0.25
gemma-4-12B-it	SD3	0.2492	0.1631	0.4	1.0	0.04
gpt-oss-20b	SD3	0.1853	0.1639	1.6	1.0	0.43
Qwen2.5-7B	SD3	0.2962	0.1665	0.3	1.0	0.05
gemma-4-12B-it	SD3	0.2526	0.1692	0.4	0.0	0.00

Table 8: Answer-equivalent for all cells. m^* is the integer crossing, m_{interp}^* its interpolation between the bracketing budgets.